Understanding gene regulation for genetic engineering

Understanding gene regulatory principles (Systems Biology) and applying gene regulatory principles for circuit design (Synthetic Biology)

Lecture 1

Somenath Bakshi

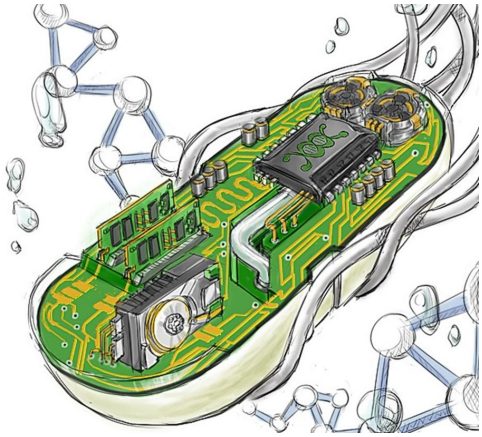
Laboratory of systems and Synthetic Microbiology

Information and Bio - engineering division, Department of Engineering

Genetic engineering and Gene editing

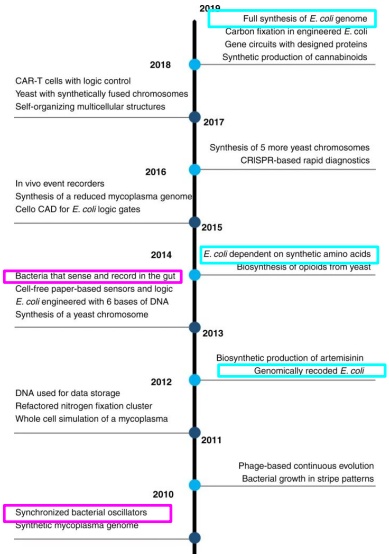
In **genetic engineering**, scientists typically stitch together long stretches of DNA and insert them into an organism's genome. These synthesized pieces of DNA could be genes that are found in other organisms, or they could be entirely novel.

*In **genome editing**, scientists typically use tools to make smaller changes to the organism's own DNA. Genome editing tools can also be used to delete or add small stretches of DNA in the genome.*



"**Synthetic biology** is a field of science that involves redesigning organisms for useful purposes by engineering them to have new abilities."

Synthetic Biology – Past, Present, and Future



The second decade of synthetic biology: 2010–2020 – by Meng and Ellis

1. "Burgers that bleed" by impossible foods
96% less land and 89% fewer greenhouse gases
available in >30,000 restaurants and 15,000 groceries

"Some plant roots "bleed" when cut. The yeast *Pichia pastoris* was engineered to produce soy leghemoglobin, which improves meaty flavours and aromas when added to a plant-based burger"

4. PROVEN, a biological nitrogen fertilizer for corn
reduces the need for chemical fertilizer by 25 lbs/acre while increasing yields by 5.8 bushels
In 2020, PROVEN was used on 250,000 acres, to be expanded to millions in 2021.

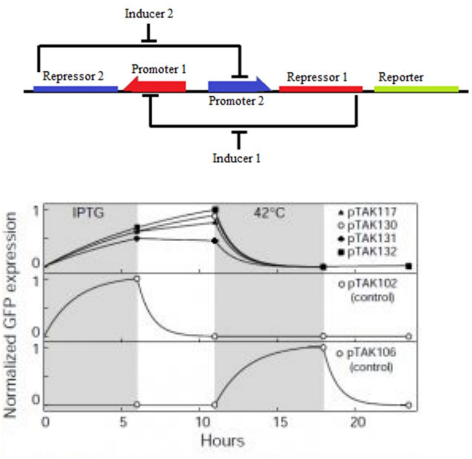
6. Calyno, a high-oleic oil from soybeans by Calyxt
contains 80% oleic acid (unmodified soy has 20%).
Calyno oil was launched in 2019 and the genome edited soy is now grown on ~100,000 acres



Synthetic biology 2020-2030 - by Chris Voight

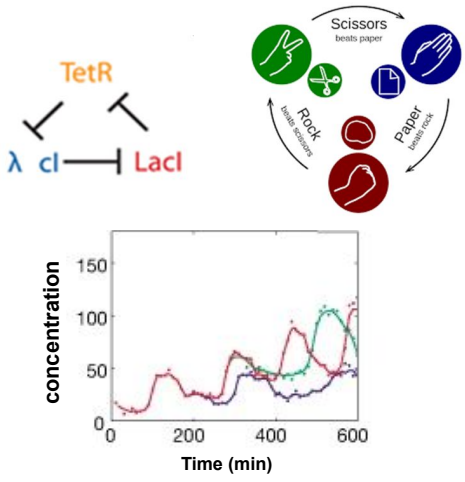
First decade of synthetic biology - clocks and switches

Genetic toggle switch



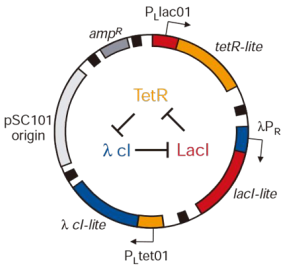
Gardner, Cantor and **Collins**, *Nature*, 2000

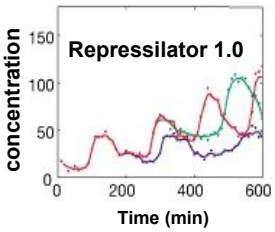
Genetic oscillator

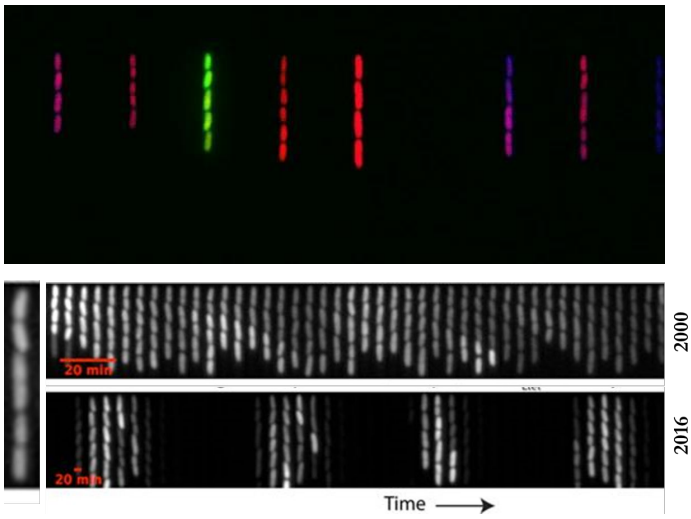


Elowitz and Leibler, *Nature*, 2000

Repressilator 2.0 and sensing in the gut

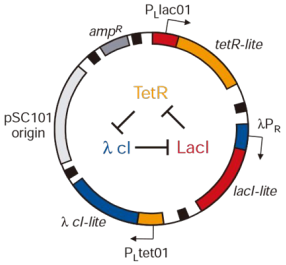


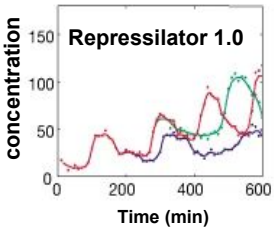


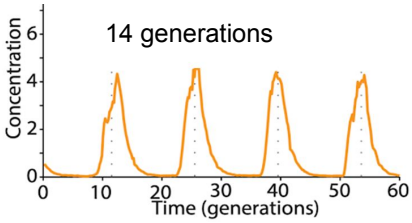


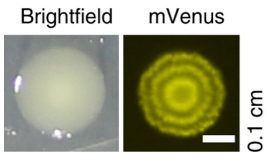
Potvin et al., *Nature*, 2016

Repressilator 2.0 and sensing in the gut

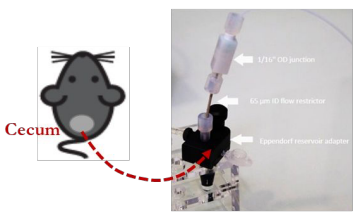


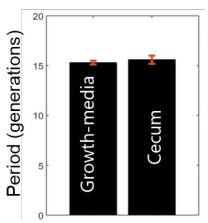


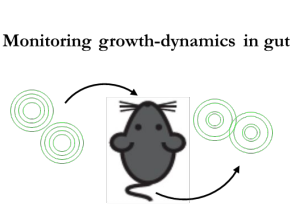




Potvin et al., *Nature*, 2016







Riglar et al., *Nature Communications*, 2019

Steps in genetic engineering

In **genetic engineering**, scientists typically stitch together long stretches of DNA and insert them into an organism's genome.

1. stitch together **long stretches of DNA**

and

2. insert them into an organism's genome.



Steps in genetic engineering

In **genetic engineering**, scientists typically stitch together long stretches of DNA and insert them into an organism's genome.

1. stitch together **long stretches of DNA**

and

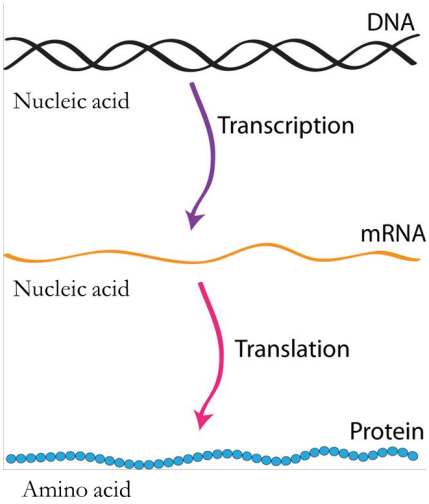
2. insert them into an organism's genome.

How do we **design** these long pieces of DNA ?

Gene regulation

(regulating gene-expression)

Gene-expression: revisiting the Central Dogma



Information storage

Gene- unit of information

1 gene – 1 message

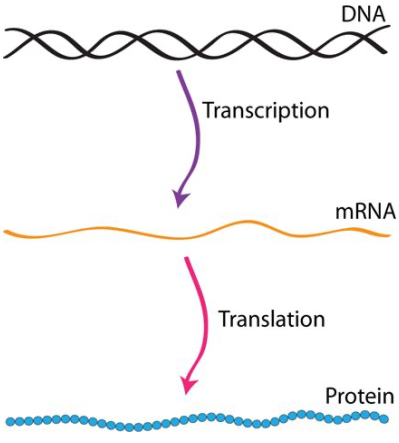
Temporary (intermediate)

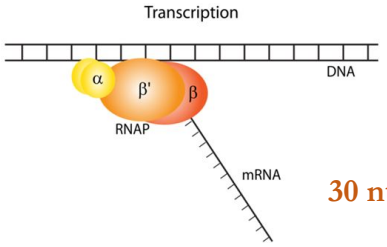
Messenger RNA

Structural and functional units of the cell

Kinetics of gene-expression

Average length of a gene : 1000 bp

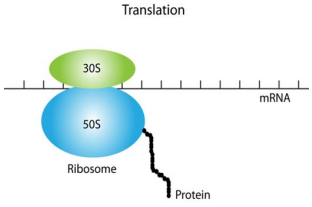




30 nt/s

Time to synthesize a typical mRNA ~ 33 sec

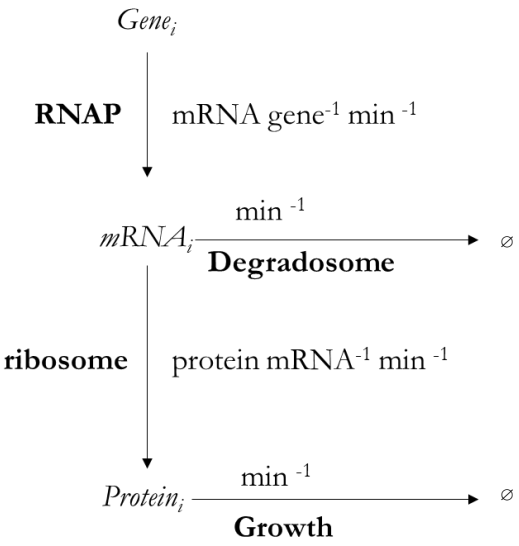
Time to synthesize a typical protein ~ 33sec



Rate of translation
10 amino acids/s

https://bionumbers.hms.harvard.edu/search.aspx

Kinetics of gene-expression



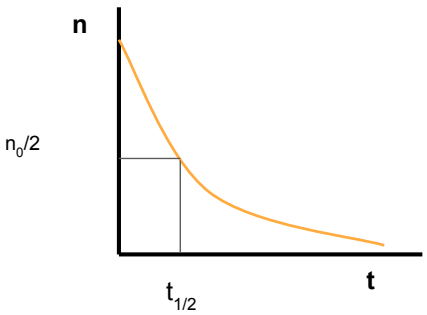
$$\frac{dm}{dt} = k_1 - \lambda_1 m \qquad (1)$$

k_1 = rate of mRNA production
λ_1 = rate of mRNA degradation

$$\frac{dp}{dt} = k_2 - \lambda_2 p \qquad (2)$$

k_2 = rate of protein production
λ_2 = rate of protein dilution

Rate of dilution/degradation and half-life time



$n = n_0 \cdot e^{-\lambda \cdot t}$

At $t = t_{1/2}$, $n = n_0/2$

Degradation rate:
$\lambda = \log(2)/t_{1/2}$

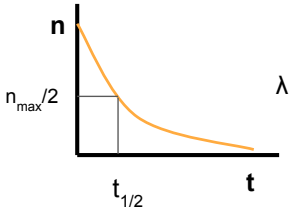
For mRNA, $t_{1/2}$ = 5 mins
$\Rightarrow$ $\lambda_1 = 0.14$ min $^{-1}$

For protein, $t_{1/2}$ = 20 mins (e.g for bacteria in rich media)
$\Rightarrow$ $\lambda_2 = 0.035$ min $^{-1}$

Estimating the steady state abundance of proteins and mRNAs

Time to transcribe an average mRNA ~ 30 sec

Time to transcribe an average protein ~ 30 sec



$\lambda = \log(2)/t_{1/2}$

For mRNA, $t_{1/2}$ = 5 mins

$\Rightarrow \lambda_1 = 0.14\ \text{min}^{-1}$

For protein, $t_{1/2}$ = 20 mins

$\Rightarrow \lambda_2 = 0.035\ \text{min}^{-1}$

$$\frac{dm}{dt} = k_1 - \lambda_1 m \quad \Rightarrow m = \frac{k_1}{\lambda_1} \text{ at steady state}$$

$k_1 = 2$ mRNA gene^{-1} min^{-1}

$\lambda_1 = 0.14$ min^{-1}

$\Rightarrow m =$ 14.3 mRNA gene^{-1}

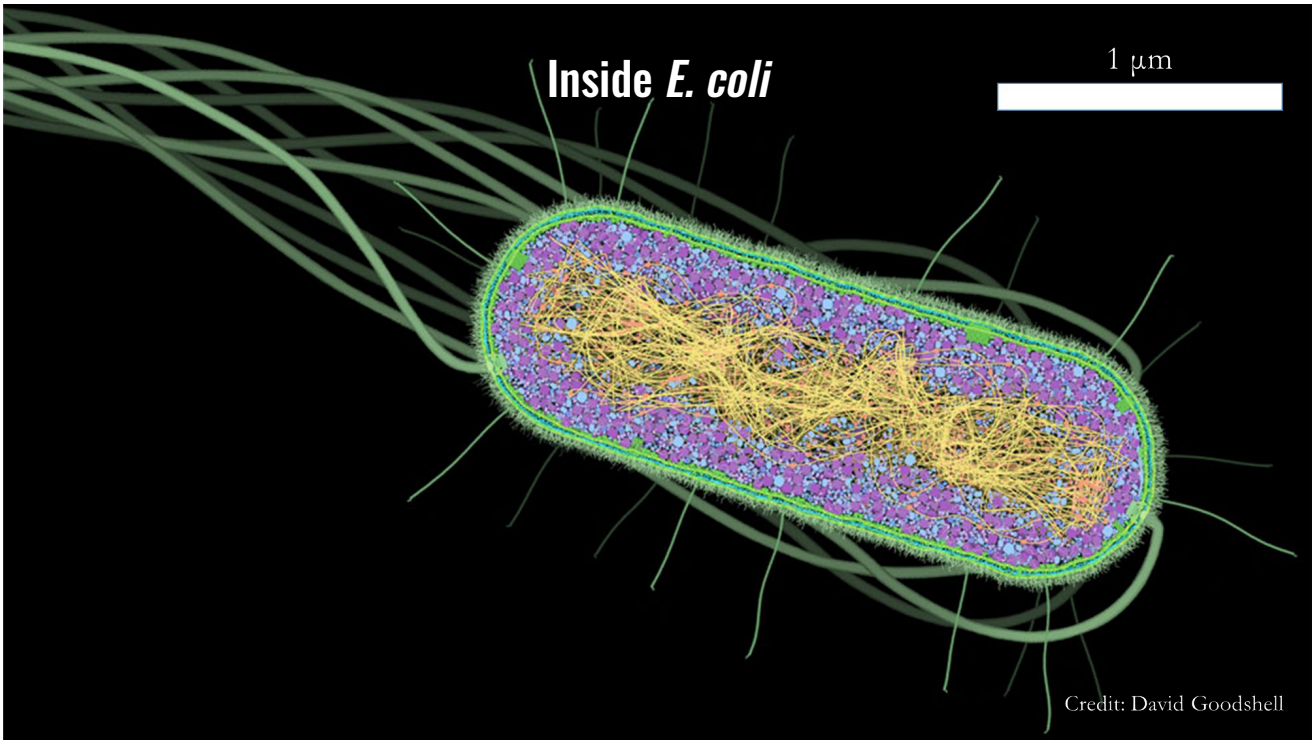
$$\frac{dp}{dt} = k_2 - \lambda_2 p \quad \Rightarrow p = \frac{k_2}{\lambda_2} \text{ at steady state}$$

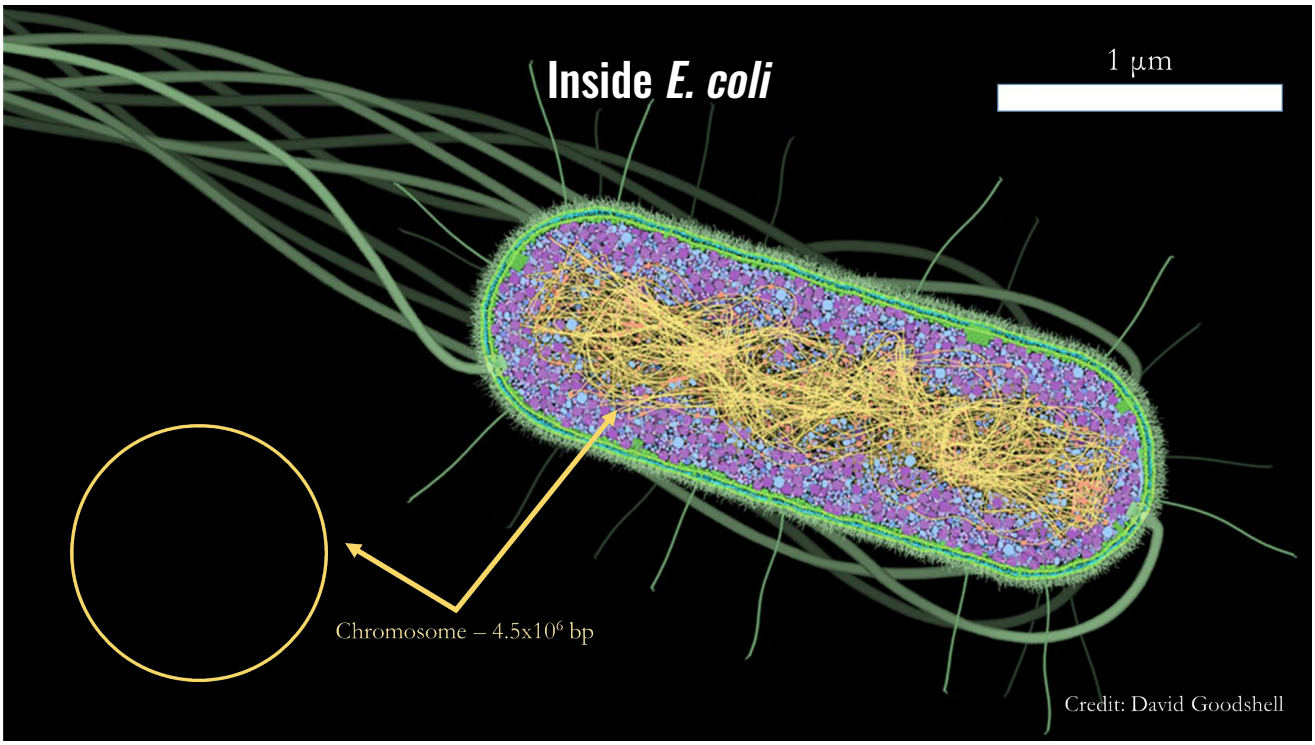
$k_2 = 2$ protein mRNA^{-1} min^{-1}

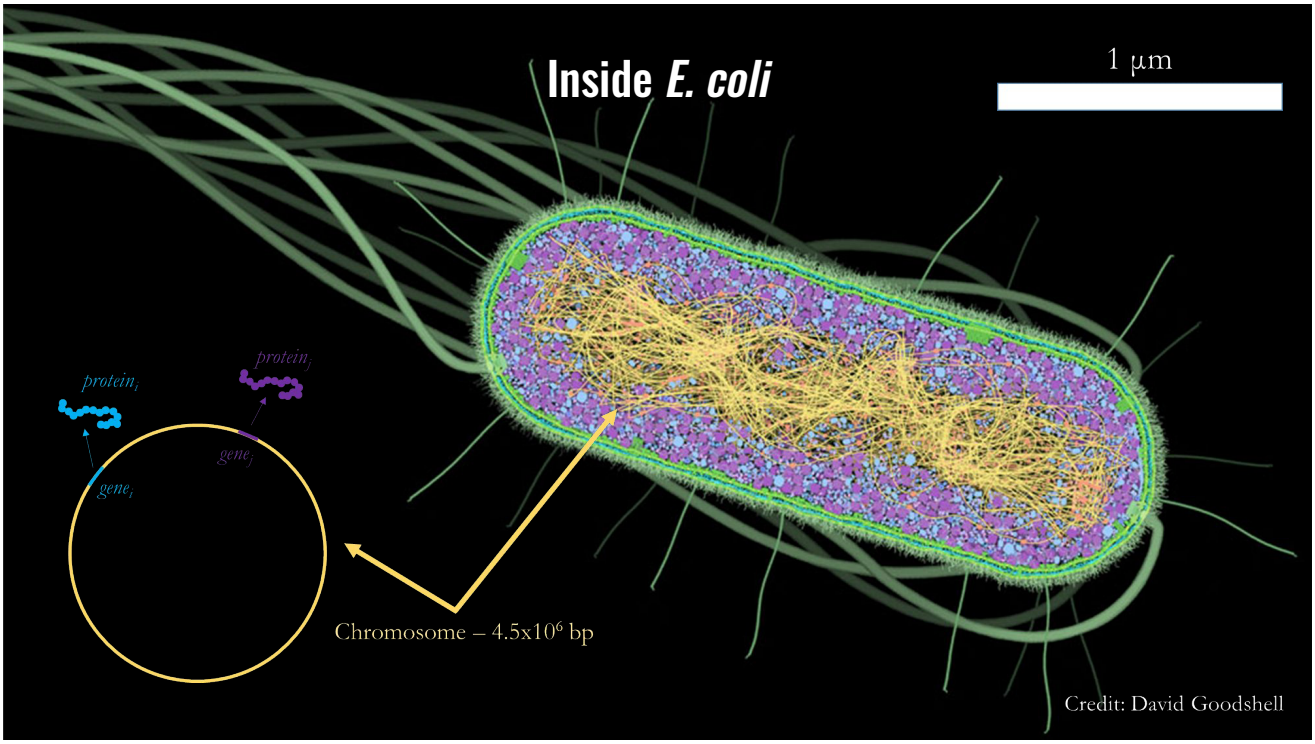
$\lambda_2 = 0.035$ min^{-1}

$\Rightarrow p =$ 57.1 protein mRNA^{-1}

= 14.3 mRNA and 816 proteins on average per gene




Inside *E. coli*
1 μm
Chromosome – 4.5x10⁶ bp
Credit: David Goodshell


Inside *E. coli*
1 μm
protein$_i$
protein$_j$
gene$_j$
gene$_i$
Chromosome – 4.5x10⁶ bp
Credit: David Goodshell

Quiz 1

If the average length of an E. coli gene is 1 kbp -

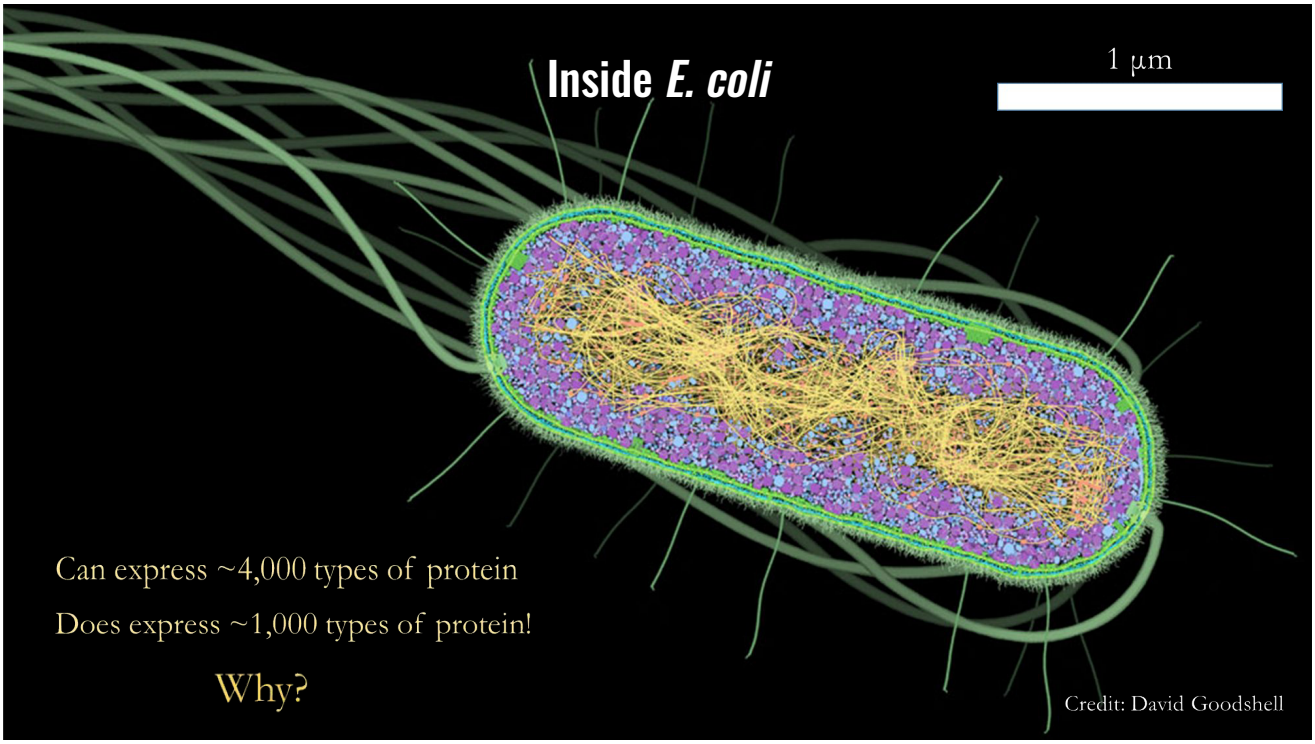
How many genes are in there in E. coli?

How many genes encode for proteins in E. coli?

EcocyC/Bionumbers

Total Genes 4494:

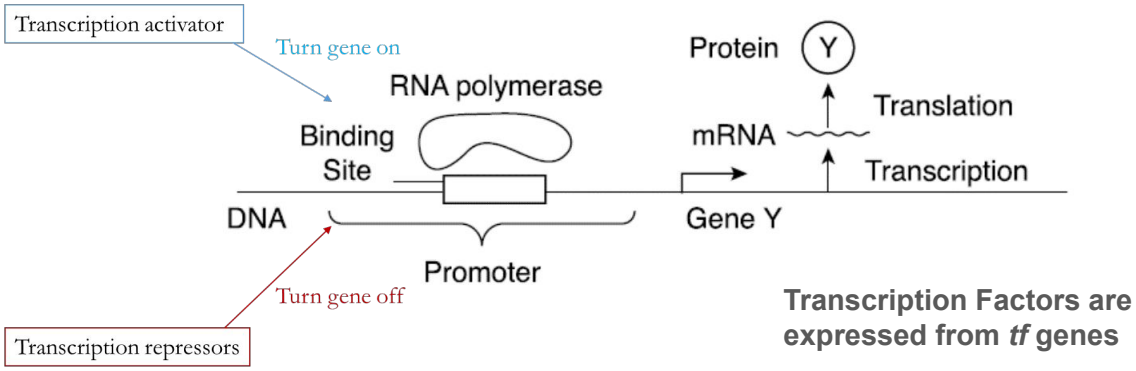
- Protein Genes 4284
- RNA Genes 198
- Pseudogenes 175



Why only 25% of the proteins are being expressed?

Why only ~25% genes (~1,000 out of ~4,000) active?

Active Genes and Transcription Factors (TF)



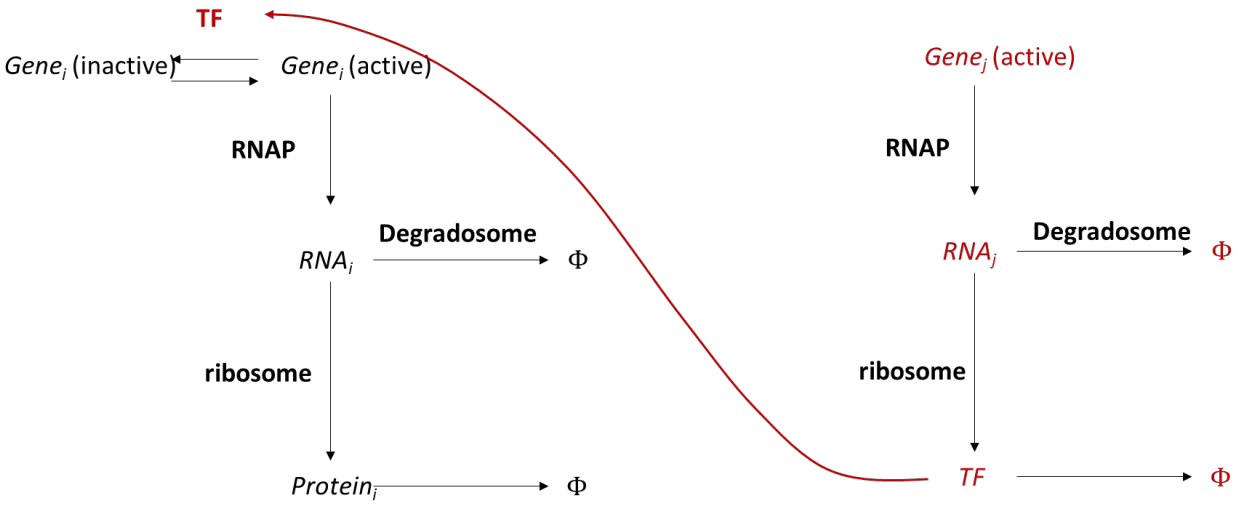
Quiz 2

Only ~25% genes (~1,000 out of ~4,000) active!

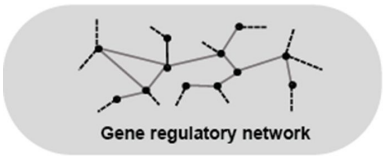
Why does the cell regulate its gene-expression?

1. Sustainability
2. Adaption
3. Development
4. All of the above
5. None of the above

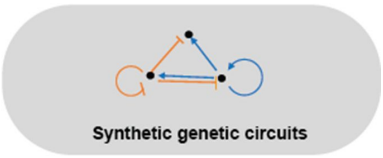
Gene regulation



Gene-regulatory networks in bacteria

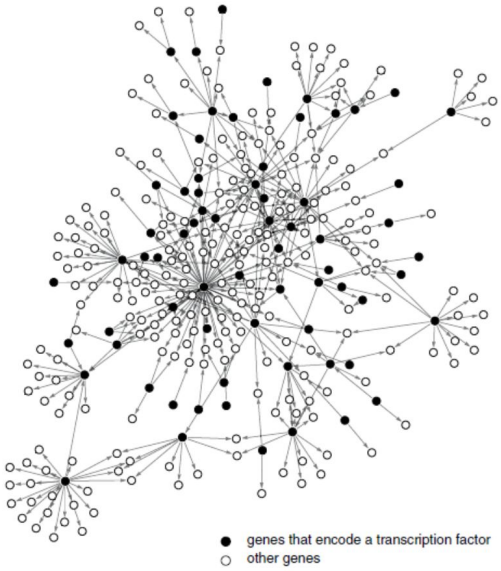


Systems Biology



Synthetic Biology

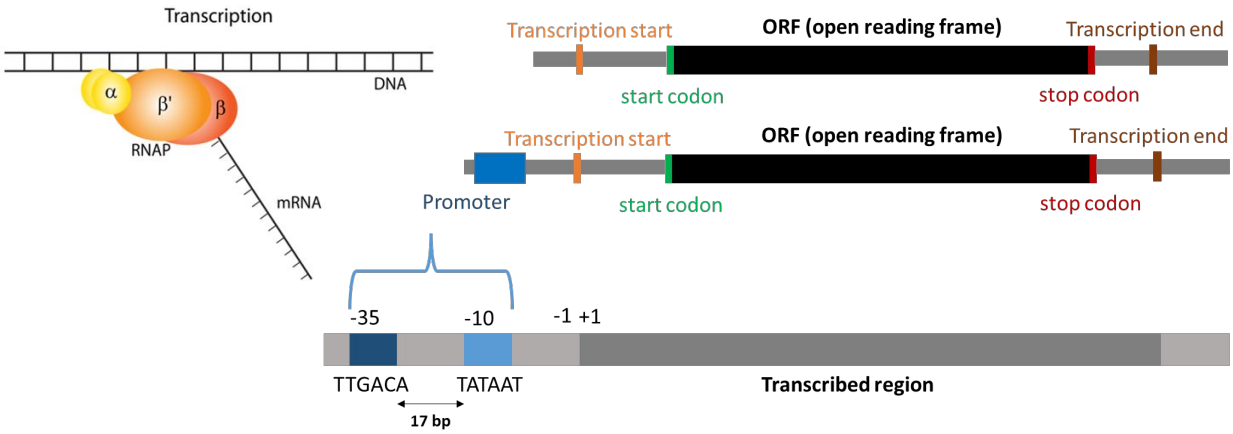
Gene-regulatory network of E. coli



- 4,400 nodes
- >10,000 edges
- 300 transcription factors

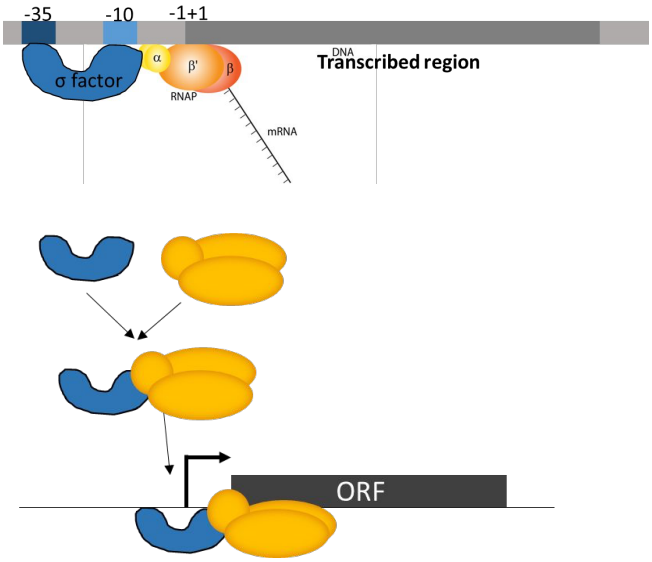
U. Alon. Intro to Systems Biology

Gene expression - transcription



The promoter sequence controls binding frequency of RNAP and differentiates between classes of genes!

Global regulation of transcription in Bacteria



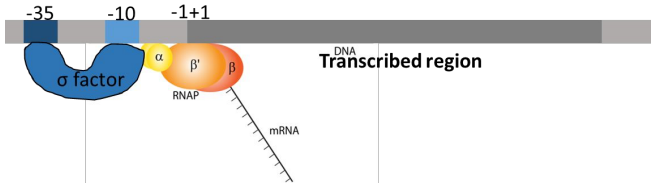
7 sigma factors

Aliases	Function	Wt. (kDa)
σ^{70}	Household sigma factor	70 ± 0.06
σ^{38}, σ^{S}	Stationary phase	38 ± 1.2
σ^{32}, σ^{H}	Heat shock (cytoplasm)	32 ± 0.02
σ^{24}, σ^{E}	Heat shock (periplasm)	22[a]
σ^{54}, σ^{N}	Nitrogen regulation	54 ± 0.2
σ^{28}, σ^{F}	Flagella biosynthesis	28 ± 0.3
σ^{19}	Ferric citrate transport	19[a]

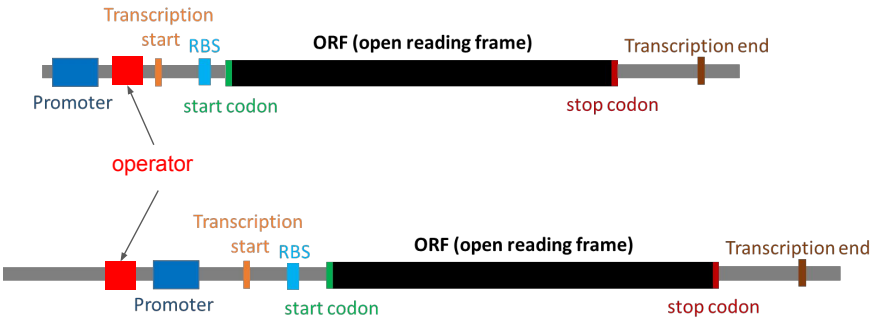
σ Factor	Promoter Consensus Sequence	
	-35 Region	-10 Region
σ^{70}	TTGACA	TATAAT
σ^{32}	TCTCNCCCTTGAA	CCCCATNTA
σ^{28}	CTAAA	CCGATAT

https://parts.igem.org/Promoters/Catalog

Gene regulation – beyond sigma factors

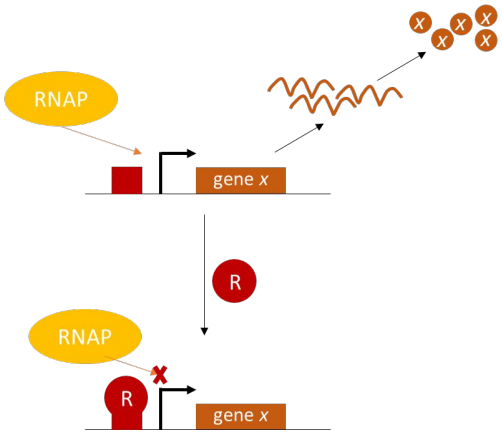


Broadly dictates which subset of genes are expressed at a given condition



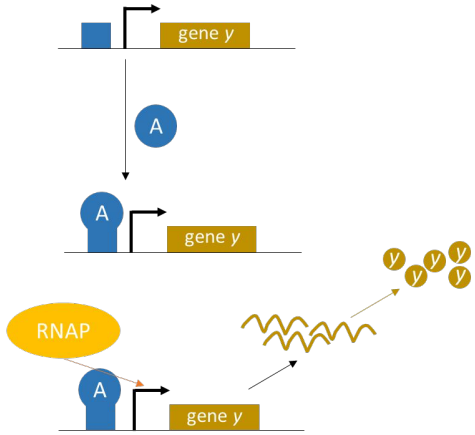
Transcriptional regulation – beyond sigma factors

Transcription repressor – turning genes off



Blocking access to the promoter

Transcription activator – turning genes on



Increasing binding probability to the promoter

Transcriptional regulation – signs of arrows

Transcription repressor – turning genes off

R ⊣ X

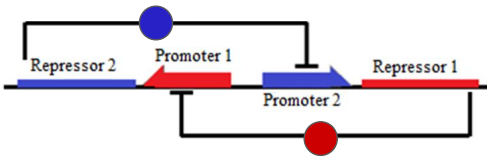
Transcription activator – turning genes on

A → Y

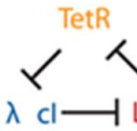
Synthetic circuits: Repressilator

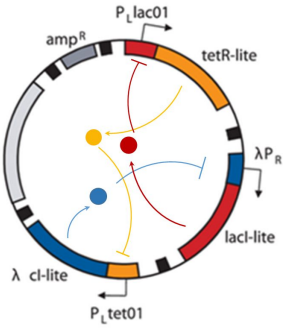
Toggle switch





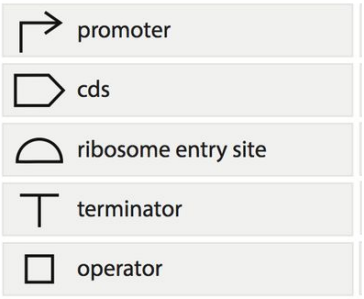
repressilator

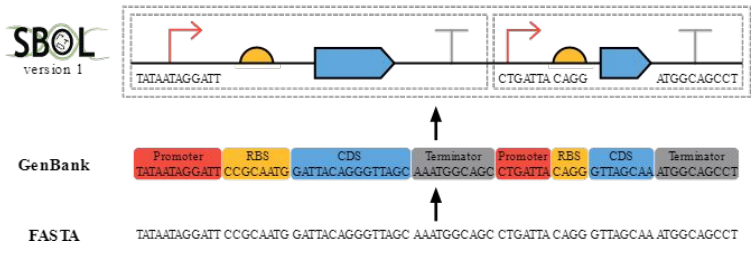


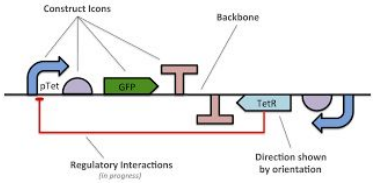


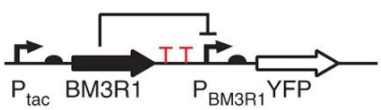
Synthetic Biology Open Language (SBOL)

open-source standard for the representation of biological designs

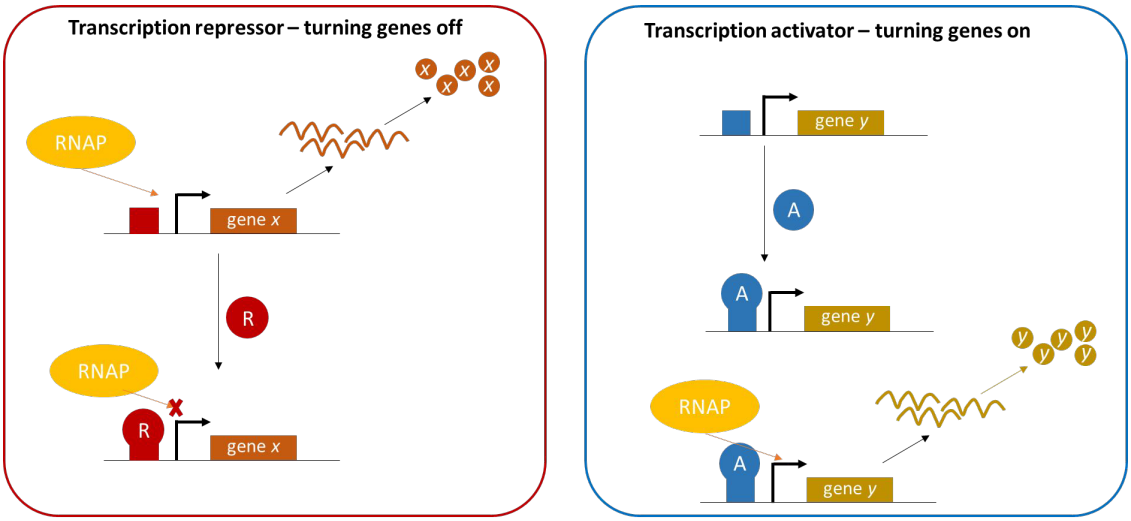






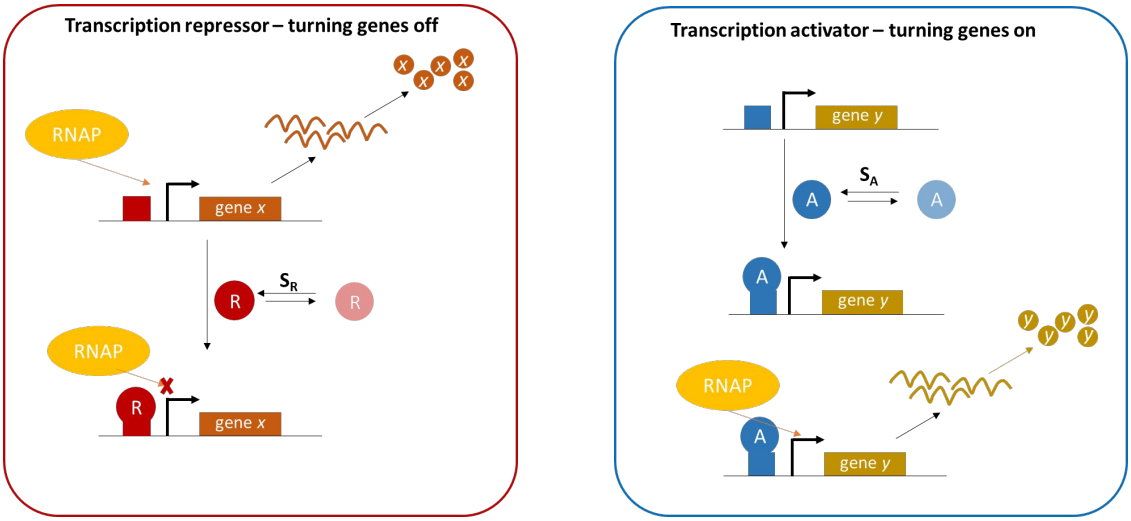


Regulation of transcription factors

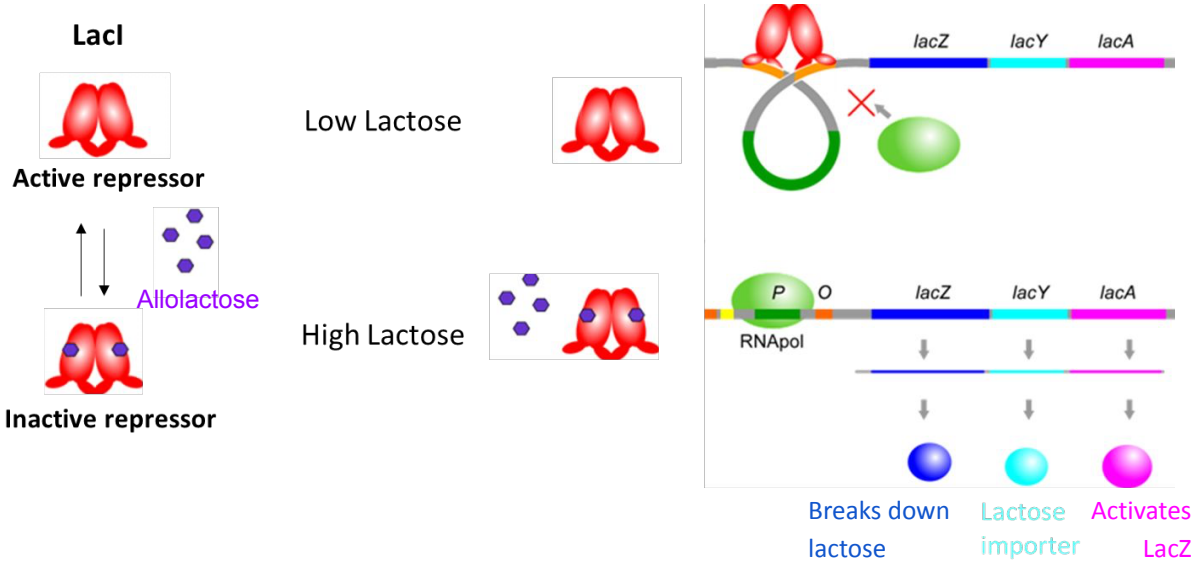


What factor regulates the transcription factor?

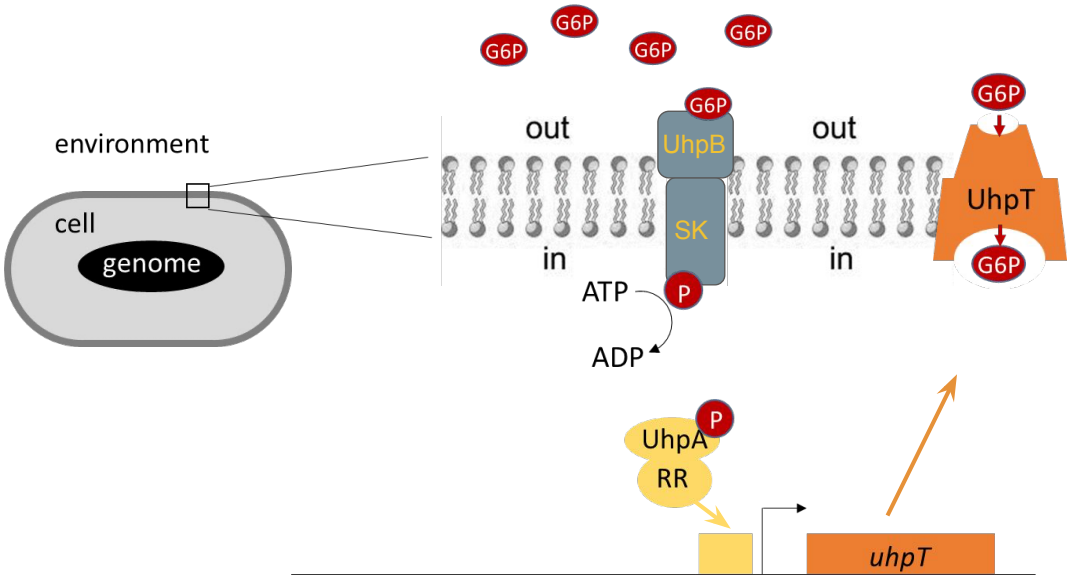
Transcriptional regulation – signal and allostery



Allostery and regulation of lac operon



Signal transduction: Intake of hexose sugar



Positive and negative control of expression



Positive control

S	A	X
0	0	0
1	1	1

Negative control

S	R	X
0	1	0
1	0	1

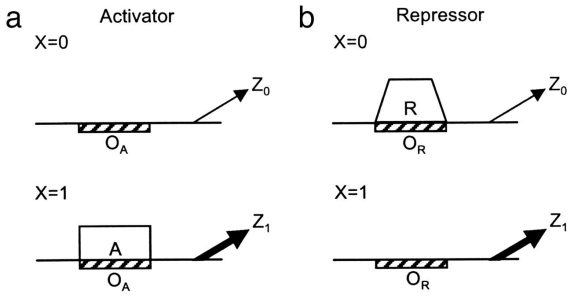
Rules for biological regulation based on error minimization

- Why does the cell needs two types of control?
- Which genes are positively controlled and which ones are negatively controlled?
- Which one should you use in your own experiments?

The demand rule:
- High-demand genes tend to have positive control
- Low demand genes tend to have negative control

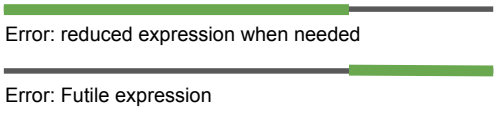
Demand: The fraction of time in the natural environment that the gene product is needed near the high end of its regulatory range

Savageu, PNAS 1977



Shinar et al., PNAS 2006

- Free sites are more error prone and exposed to binding from nonspecific factors



Error: reduced expression when needed

Error: Futile expression

Summary

1. Cells only express a subset of the genes under a given environment. Transcription factors are special proteins that regulate which proteins are expressed.

2. The promoter sequence (strength) determines the transcription frequency of a gene and the RBS sequence (strength) determines the translation frequency of an mRNA

3. Sigma factors are global transcription factors that determine which subset of genes can be expressed at a given condition. Specific transcription factors regulate specific genes within these subsets.

4. Signalling molecules can regulate the state of transcription factors through allostery, to activate or inactive them.

5. Gene expression can be regulated by positive and negative modes of control through the signal molecule and in nature the choice is dictated by the demand of that gene.